Summary - Modelling of Liquid-liquid phase separation in peptides

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1 Modelling

The model includes a total of 5 species, which we track by the volume fractions of the different components:

ϕ_w	water,
ϕ_+	salt cations (Na^+ or Mg^{2+}, charge $z_+ = +1$ or $+2$, respectively),
ϕ_-	salt anions (Cl^-),
ϕ_p	peptide A molecules (charge $z_p = +2$).
ϕ_c	counterions (OH^-).

The volume fraction ϕ_i is linked to the number density of a species n_i via $\phi = \nu_i n_i$, where ν_i is the molecular volume of species i. For simplicity, and in line with previous work, we assume that solvent molecules, salt ions have an equal molecular volume $\nu = 0.018$ l/mol. The monomers of the petides, that is, the amino-acides, are roughly 7 times larger. Since each peptide used in the experiments has 7 amino acids, we assume that the molecular volume of the peptides is $\nu_p = \bar{\nu}_p \nu$, with $\bar{\nu}_p = 50$. Thus $\nu_p = 0.9$ l/mol.

We assume that the five species fill up the entire space, i.e. the volume fractions satisfy the no-void condition:

$$\phi_+ + \phi_- + \phi_w + \phi_c + \phi_p = 1, \tag{1}$$

where $\nu_p = \nu N_m$, where N_m is the number of the monomers. We further assume that the solution is overall electroneutral (*i.e.*, there is no imposed electric potential, so the total charge must be zero)

$$z_+\phi_+ - \phi_- - \phi_c + \frac{z_p}{\bar{\nu}_p}\phi_p = 0, \tag{2}$$

where the first two terms capture the net charge of dissociated ions in solution ($z_+ = 1, 2$ depending on the salt considered), while the last term captures the net charge of ions associated with the peptides that has a total charge $z_p = 2$.

Free energy

We consider the mixture to be incompressible and kept at some constant temperature T. Then, the electro-chemo-thermodynamics of the mixture is governed by the Helmholtz free-energy f, which consists of three terms [1, 2]

$$f = f_{\text{FH}} + f_{\text{DH}}, \tag{3}$$

which includes the Flory-Huggins free energy for the demixing of multiphase fluids and the Debye-Huckel free energy that captures electrostatic fluctuations. In defining the Flory-Huggins free energy, we account for the entropy of each component and hydrophobic interactions between the peptides and the solvent.

$$\frac{\nu f_{\text{FH}}}{k_B T} = \sum_{i=\pm,w,c} \phi_i \ln(\phi_i) + \phi_p \ln(\phi_p) + \chi \phi_p n_w. \tag{4}$$

We note that χ could be taken to be a function of the charge state of the peptide and solvent, but here we instead consider the case in which short-range hydrophobic interactions are independent of the charge state. However, in the absence of detailed functional forms, we here consider the simplest case in which χ, and therefore hydrophobic interactions, are independent of the charge state of the components. Finally, we account for electrostatic fluctuation in electroneutral solutions by including the Debye-Hückel electrostatic term f_{DH} [1, 2]

$$\frac{\nu f_{\text{DH}}}{k_B T} = -\frac{1}{4\pi}\left(\log(1+\kappa) - \kappa + \frac{\kappa^2}{2}\right) \tag{5a}$$

where κ satisfies

$$\kappa^2 = \lambda\left(z_+^2\phi_+ + \phi_- + \frac{z_p}{\bar{\nu}_p}\phi_p + \phi_c\right). \tag{5b}$$

Note that in writing Eq. (5b), we have assumed that the charges on the peptides can be treated analogously to free ions. An alternative would be to use

$$\kappa^2 = \lambda\left(z_+^2\phi_+ + \phi_- + \frac{z_p^2}{\bar{\nu}_p}\phi_p + \phi_c\right). \tag{5b*}$$

The λ constant in (5b) is a physical non-dimensional scale measuring the ratio between the Bjerrum length and the characteristic size of ions, $\lambda = 4\pi\ell_B/\sqrt[3]{\nu}$. We will use it as control parameter.

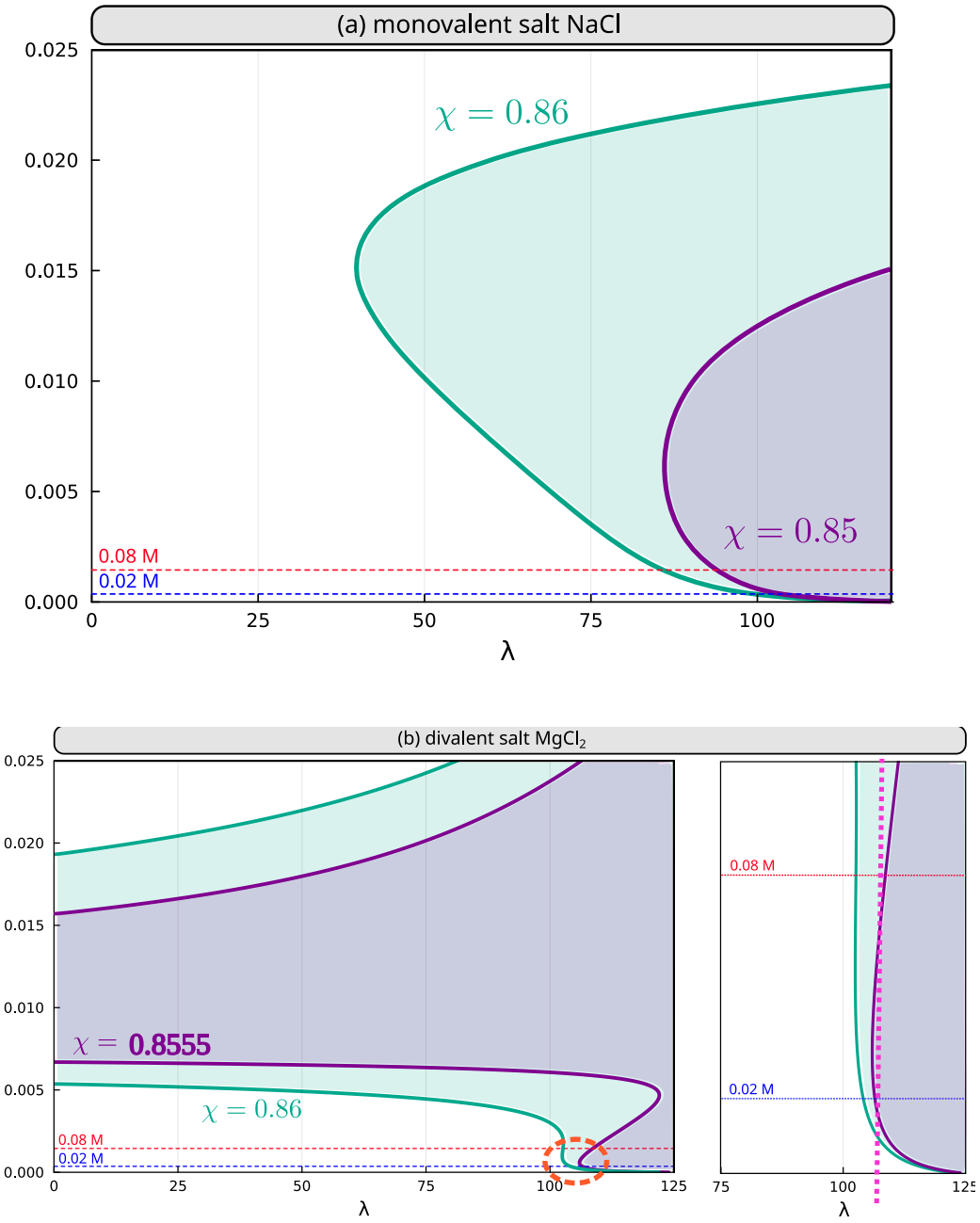


Figure 1: For any value of λ the range of salt concentrations for which the system demix is highlighted by the shaded region. The vertical axis is the total volume fraction of cations, that is $\phi_+^{(0)}$. For comparison, we indicate the range of salt concentration used experimentally and that delineate demixing when divalent salt is used.

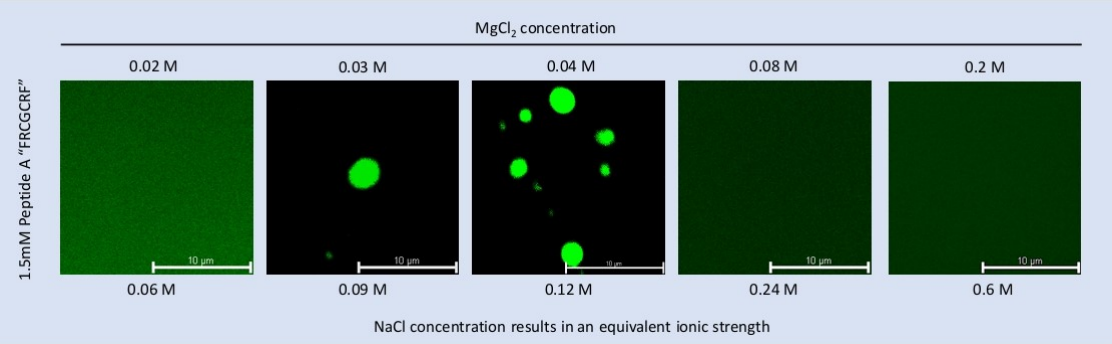


Figure 2: Preliminary experimental results using $NaCl/MgCl_2$.

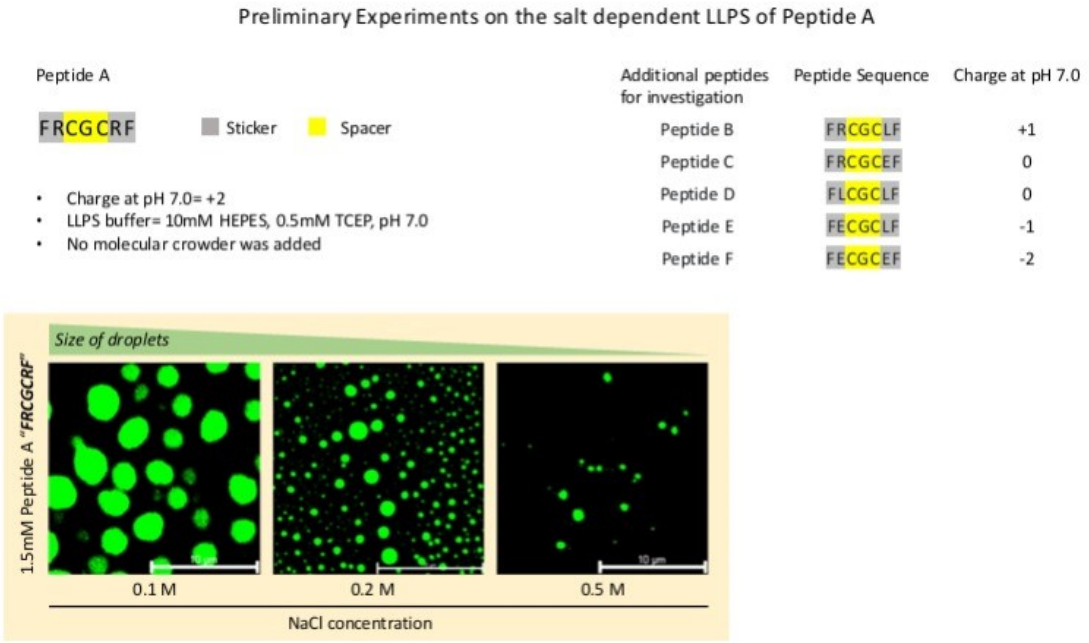


Figure 3: Experimental results for Peptide A.

Phase separation

Conditions for the coexistence of multiple homogeneous phases, each characterised by

$$\boldsymbol{\phi}^{\mathsf{I}/\mathsf{II}} = (\phi_+^{\mathsf{I}/\mathsf{II}}, \phi_-^{\mathsf{I}/\mathsf{II}}, \phi_p^{\mathsf{I}/\mathsf{II}}, \phi_c^{\mathsf{I}/\mathsf{II}}, \phi_w^{\mathsf{I}/\mathsf{II}}),$$

are obtained by minimising the combined free energy of the two phases

$$r f(\boldsymbol{\phi}^{\mathsf{I}}) + (1-r) f(\boldsymbol{\phi}^{\mathsf{II}}) = \min! \tag{6}$$

Here, r and $(1-r)$ are the proportions of a unit size volume taken up by phase (I) and (II), respectively. The value of r is in general an unknown that is part of the minimisation process, and is fixed by the composition of the well-mixed material occupying a unit volume. The minimum needs to be taken under several constraints:

(a) The no-void (1) and the electro neutrality conditions (2) have to be satisfied in each of the two phases separately.

(b) The total amount of each species must be equal to the fraction of this species in the well-mixed state:

$$r\phi_i^{\mathsf{I}} + (1-r)\phi_i^{\mathsf{II}} = \phi_i^{(0)}, \tag{7}$$

for all five species $i = +, -, c, p, w$. The superscript (0) indicates the volume fraction of the species in the well-mixed state. However, two of these conditions are redundant (as they are implied by the no-void and the electroneutrality conditions). So we only retain the conditions for $i = +, -, p$.

Using Lagrange-multipliers for these conditions gives 18 variables, and correspondingly 18 necessary conditions. For each choice of $\phi_\pm^{(0)}$ and $\phi_p^{(0)}$, we obtain one solution, suggesting a three dimensional parameter space $\mathbb{U}_0 \subset [0,1]^3$ to explore. Basically, once all the material parameters (χ, λ, etc.) are set, one chooses a point $(\phi_+^{(0)}, \phi_-^{(0)}, \phi_p^{(0)})$ which characterises the homogeneous mixture of the material prior to phase separation. Then solving the 18 equations gives the ten volume fractions characterising the two phases $\boldsymbol{\phi}^{\mathsf{I}/\mathsf{II}}$, the proportion r of the two phases, and a bunch of Lagrange parameter values, *provided* the mixture permits separation into two phases. If not, then only the trivial solution for the homogeneous mixture exists ($\boldsymbol{\phi}^{\mathsf{I}} = \boldsymbol{\phi}^{\mathsf{II}}$). So we can split the three dimensional space $\mathbb{U}_0$ into (the 3D representation of) the coexistence region $\mathbb{U}$ where phase separation is possible and $\mathbb{U}_0 \backslash \mathbb{U}$ where only the homogeneous solution is available. The boundary is characterised by $r = 0$ or $r = 1$ i.e. where the homogeneous mixture is one of the pure phases. Hence we can restrict our problem to finding surfaces of the three dimensional

coexistence region $\mathbb{U}$. Moreover, we will only consider cases for which the total amount of + and − species (salt cations and anions) satisfy the stoichiometry of the salt.

In essence, we are now looking for boundary curves of a two-dimensional subset of $\mathbb{U}$. Working out the implications of this allows us to remove r from the problem (we effectively set $r \to 1$), so that we now have 17 equations and instead of the three conditions (7) we only have two, which are `f[16]` and `f[17]` in Andreas' code. Only $\phi_p^{(0)}$ remains of the five parameters for the composition of the homogeneous mixture, labelled as `KM` in the code.

The result of these calculations is shown in fig. 1. The top figure, fig. 1(a), is for the monovalent salt (sodium chloride) and the bottom one fig. 1(b) for the divalent salt. The calculations were done with a fixed total amount of peptide A, namely 1.5 mM (i.e. volume fraction of $\phi_p^{(0)} = \texttt{KM} = 1.35 \times 10^{-3}$), and two choices of χ. The figures show the region of total cation volume fraction $\phi_+^{(0)}$ on the vertical axis vs. the value of λ on the horizontal axis. There are also two dashed blue and red lines lines indicating the volume fractions for 0.02M and 0.08M, corresponding to 0.36×10^{-3} and 1.44×10^{-3} respectively. These are the range of values used in the experiments where phase separation is seen for the divalent salt, see fig. 2 from IIT's experiments. For the monovalent salt, phase separation was observed in all cases in fig. 3 and the range of total cation concentration reported in that figure was from 0.1M to 0.5M, or correspondingly a volume fraction $\phi_+^{(0)}$ from 1.8×10^{-3} to 9×10^{-3}.

Focussing first on the top figure for the monovalent salt, and the smaller value of $\chi = 0.85$, the values of salt for which phase separation occurs is located between the upper and lower part of the violet curve and is shaded grey. For the larger value $\chi = 0.86$, the region (the area between the parts of the green curve) extends further to the left i.e. to smaller value of λ. None of these regions include the $\lambda = 0$ value, that is, the Debye-Hückel effect is crucial here for the instability (at least for these ranges of χ). Importantly, the within the range of unstable range of salt concentrations, there is at most a lower stability transtion, but never at the top range. That is consistent with the observation that phase separation persists even for the highest experimentally explored salt concentrations.

For the divalent salt, fig. 1(b), the regions extend to the left to $\lambda = 0$ and are generally larger. So phase separation is more prevalent with respect to this parameter. Interestingly, there is an 'S' shape in the line for $\chi = 0.855$, indicated by a red dashed ellipse. An enlarged version of this region is shown in the right half of figure (b). If we fix λ to about 105, we see that the vertical red dashed line crosses the violet line twice. This means that for low values of salt (total cation

concentration below about 0.02M), the components would mix, and also for values above about 0.08M. Between these two values, there would be phase separation. So this agrees with the experimental observation in fig. 2.

References

[1] Bin Zheng, Yael Avni, David Andelman, and Rudolf Podgornik. Phase Separation of Polyelectrolytes: The Effect of Charge Regulation. *The Journal of Physical Chemistry B*, 125(28):7863–7870, 2021. Publisher: American Chemical Society.

[2] Giulia L Celora, Ralf Blossey, Andreas Münch, and Barbara Wagner. Counterion-controlled phase equilibria in a charge-regulated polymer solution. *The Journal of Chemical Physics*, 159(18), 2023.